

Question number	Answer	Mark
3(a)(i)	The only correct answer is D <i>A is incorrect because fibrinogen is in the plasma</i> <i>B is incorrect because prothrombin is in the plasma</i> <i>C is incorrect because thrombin is formed from prothrombin which is in the plasma</i>	(1)

Question number	Answer	Mark
3(a)(ii)	The only correct answer is C <i>A is incorrect because fibrinogen is not an enzyme, and prothrombin is a precursor enzyme</i> <i>B is incorrect because prothrombin is a precursor enzyme</i> <i>D is incorrect because prothrombin is a precursor enzyme</i>	(1)

Question number	Answer	Mark
3(a)(iii)	The only correct answer is A <i>B is incorrect because fibrinogen is soluble</i> <i>C is incorrect because prothrombin is soluble</i> <i>D is incorrect because thromboplastin is soluble</i>	(1)

Question number	Answer	Additional guidance	Mark
3(b)	An answer that includes three of the following points: • blood clot (in pulmonary artery) {prevents / reduces} (deoxygenated) blood reaching lungs (1) • so blood will {not become oxygenated / contain less oxygen} (1) • therefore there will {not be enough / less} oxygen supplied to the {cells / tissues} (1) • difficulty in breathing due to body trying to increase gas exchange (1) • rapid heart rate to pump more {(oxygenated) blood / oxygen} {around the body / to the cells} (1)	ACCEPT less gas exchange ACCEPT named cell / tissue e.g. heart tissue IGNORE body ACCEPT increase {oxygen uptake / removal of carbon dioxide} ACCEPT named cell / tissue e.g. lungs	(3)

Question number	Answer	Additional guidance	Mark
3(c)	An answer that includes three of the following points: • (would be useful) if we could find a way to {block / breakdown} HSP47 (1) • (reduced HSP47) would {prevent / reduce risk of} {DVT / thrombus / blood clotting} (1) • and therefore reduce risk of {pulmonary embolism / CVD / CHD} (1) • may not have the {side effects / named side effect} of current {anticoagulants / platelet inhibitors} (1) • HSP47 {may not be present in humans / may be species specific} (1)	ACCEPT {find a way to reduce HSP47 / reduce synthesis of HSP47 / modify HSP47 gene} using HSP47 to increase blood clotting process in e.g. surgery, haemophilia ACCEPT description of blood clotting process being reduced ACCEPT may cause side effects e.g. prevent blood from clotting, excessive bleeding	(3)